

RAK12010 WisBlock Ambient Light Sensor Module Datasheet

Overview

Description

RAK12010 is a WisBlock Sensor that extends the WisBlock system to have ambient light sensing capability. It uses the VEML7700 chip from Vishay Semiconductors with an I2C interface making it easy to build an ambient light data acquisition system.

Features

- Ambient Light Sensor (ALS) module
- Digital 16-bit resolution sensor
- **Ambient Light Range:** 0 lx to about 120,000 lx
- **Ambient Light Resolution:** 0.0036 lx
- Excellent temperature compensation
- 3.3 V Power Supply
- Current Consumption: < 2 - 45 uA
- Chipset: Vishay Semiconductors VEML7700
- **Module size:** 10 X 10 mm

Specifications

Overview

Mounting

Figure 1 shows the mounting mechanism of the RAK12010 module on a [WisBlock Base](#)  board. The RAK12010 module can be mounted on the slots: **A, B, C, D, E, & F.**

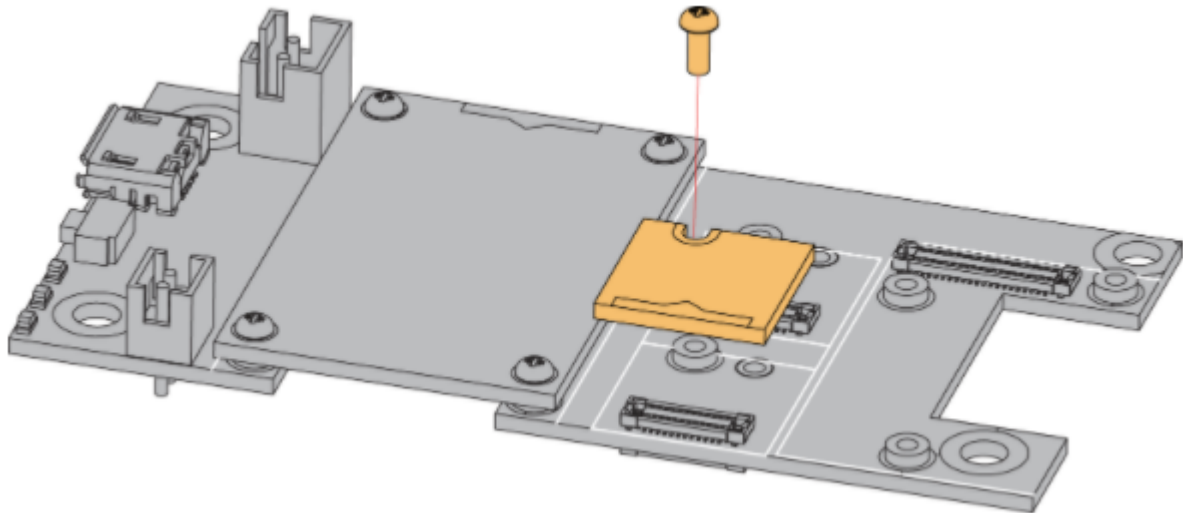


Figure 1: RAK12010 mounting mechanism on a WisBlock Base module

Hardware

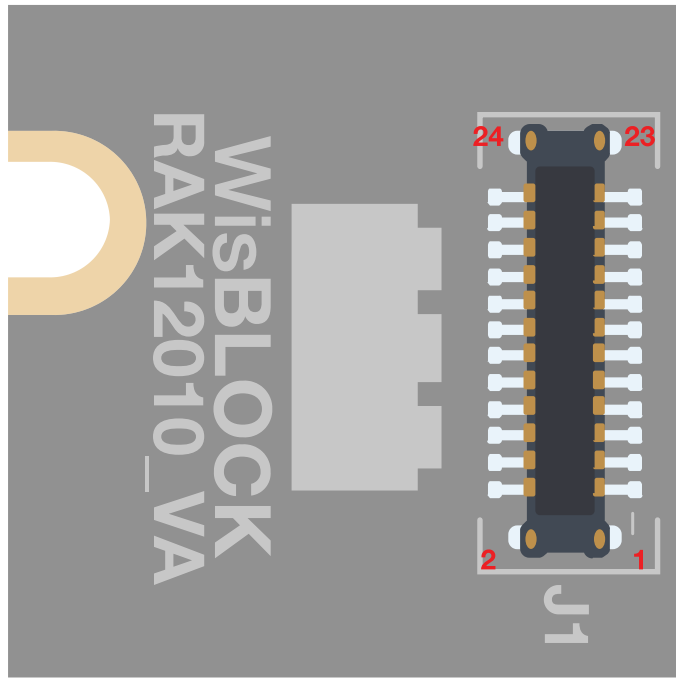
The hardware specification is categorized into five parts. It shows the chipset of the module and discusses the pinouts and its corresponding functions and diagrams. It also covers the electrical and mechanical parameters that include the tabular data of the functionalities and standard values of the RAK12010 WisBlock Ambient Light Sensor Module.

Chipset

Vendor	Part number
Vishay Semiconductors	VEML7700

Pin Definition

The RAK12010 WisBlock Ambient Light Sensor Module comprises a standard WisBlock connector. The WisBlock connector allows the RAK12010 module to be mounted to a WisBlock Base board. The pin order of the connector and the pinout definition is shown in **Figure 2**.



GND	2	1	NC
NC	4	3	NC
NC	6	5	NC
I2C SDA	8	7	I2C SCL
NC	10	9	NC
NC	12	11	3V3_S
3V3_S	14	13	NC
NC	16	15	NC
I2C SCL	18	17	I2C SDA
NC	20	19	NC
NC	22	21	NC
NC	24	23	GND

Figure 2: RAK12010 Pinout Diagram

NOTE

- Only I2C related pins, 3V3_S, and GND are connected to the WisConnector.
- 3V3_S voltage output from the WisBlock Base that powers the RAK12010 module can be controlled by the WisBlock Core via WB_IO2 (WisBlock IO2 pin). This makes the module ideal for low-power IoT projects since the WisBlock Core can totally disconnect the power of the RAK12010 module.

Electrical Characteristics

This sections shows the maximum and minimum ratings of the RAK12010 module and its recommended operating conditions. Refer to the table presented below.

Absolute Maximum Ratings

Parameter	Minimum	Maximum	Unit
VBAT	-0.3	+4.20	V
3V3_S	0	3.6	V
Total power dissipation	0	50	mW

Power Supply Ratings

Symbol	Description	Condition	Min.	Nom.	Max.	Unit
VBAT	Supply Voltage	Input voltage must within this range	3.3	4.0	4.2	V
3V3_S	VEML7700 Operating Voltage	Input voltage must within this range	2.5	3.3	3.6	V
I _{sd}	Shut down current	VDD is 3.3 V	-	0.5	-	uA
I _{DD1}	Operation mode current	VDD is 3.3 V, PSM = 11, refresh time 4100 ms	-	2	-	uA
I _{DD2}	Operation mode current	VDD is 3.3 V, PSM = 00, refresh time 600 ms	-	8	-	uA
I _{DD3}	Operation mode current	VDD is 3.3 V, PSM_EN = 0, refresh time 100 ms	-	45	-	uA

Mechanical Characteristics

Board Dimensions

The mechanical dimensions of the RAK12010 module is shown in **Figure 3** below.

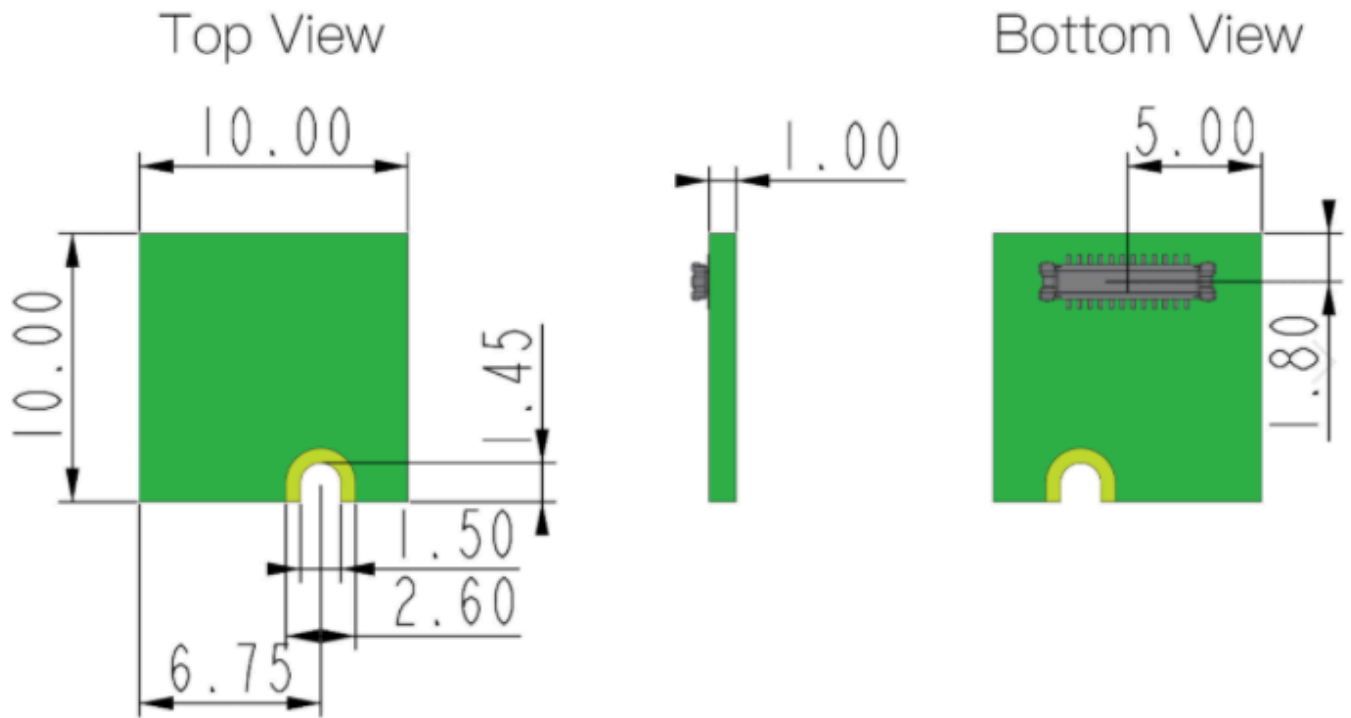


Figure 3: RAK12010 Mechanical Dimensions

WisConnector PCB Layout

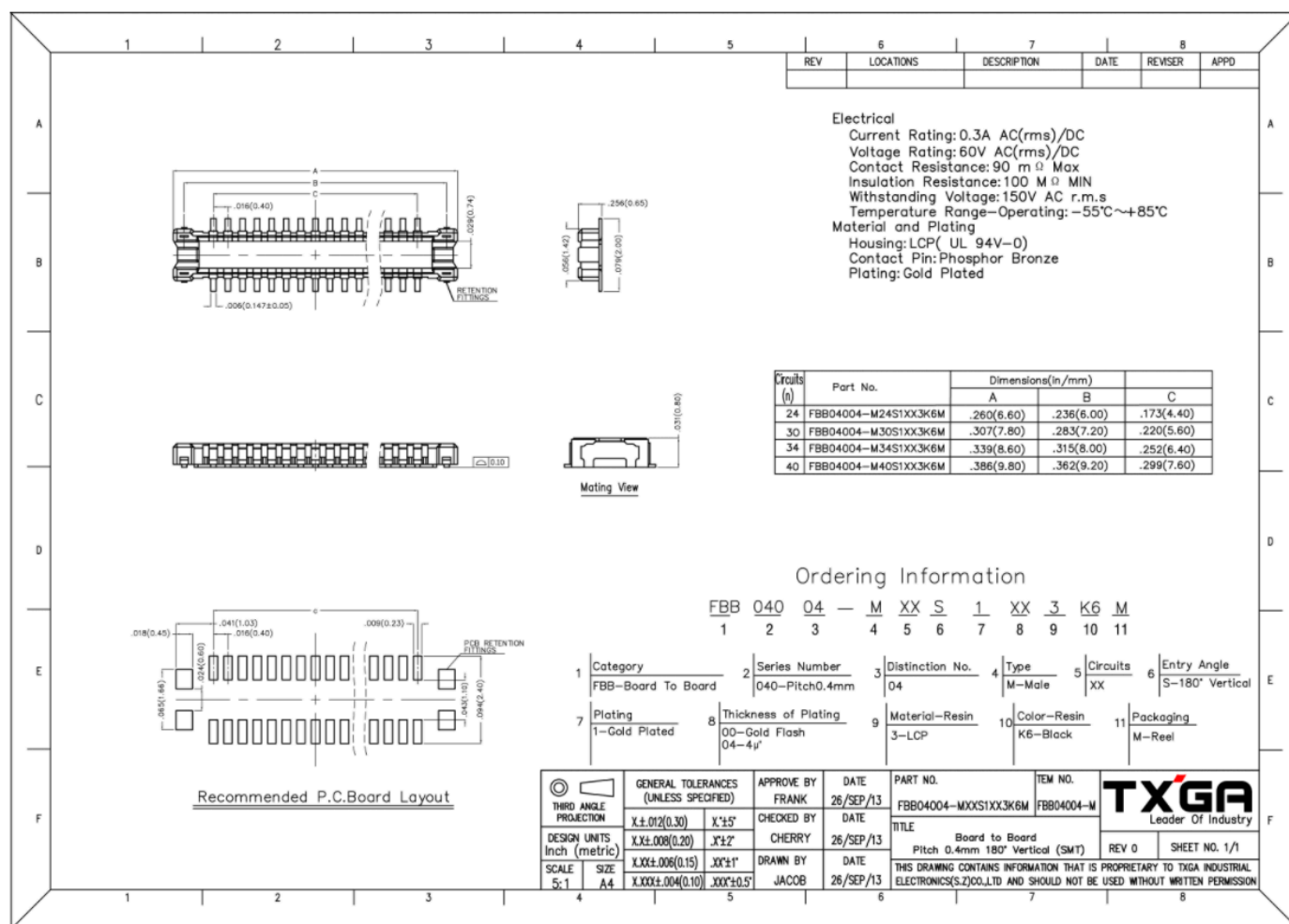


Figure 4: WisConnector PCB footprint and recommendations

Schematic Diagram

Figure 5 shows the schematic of the RAK12010 module.



Figure 6 shows the PCB Silkscreen of the RAK12010 module.

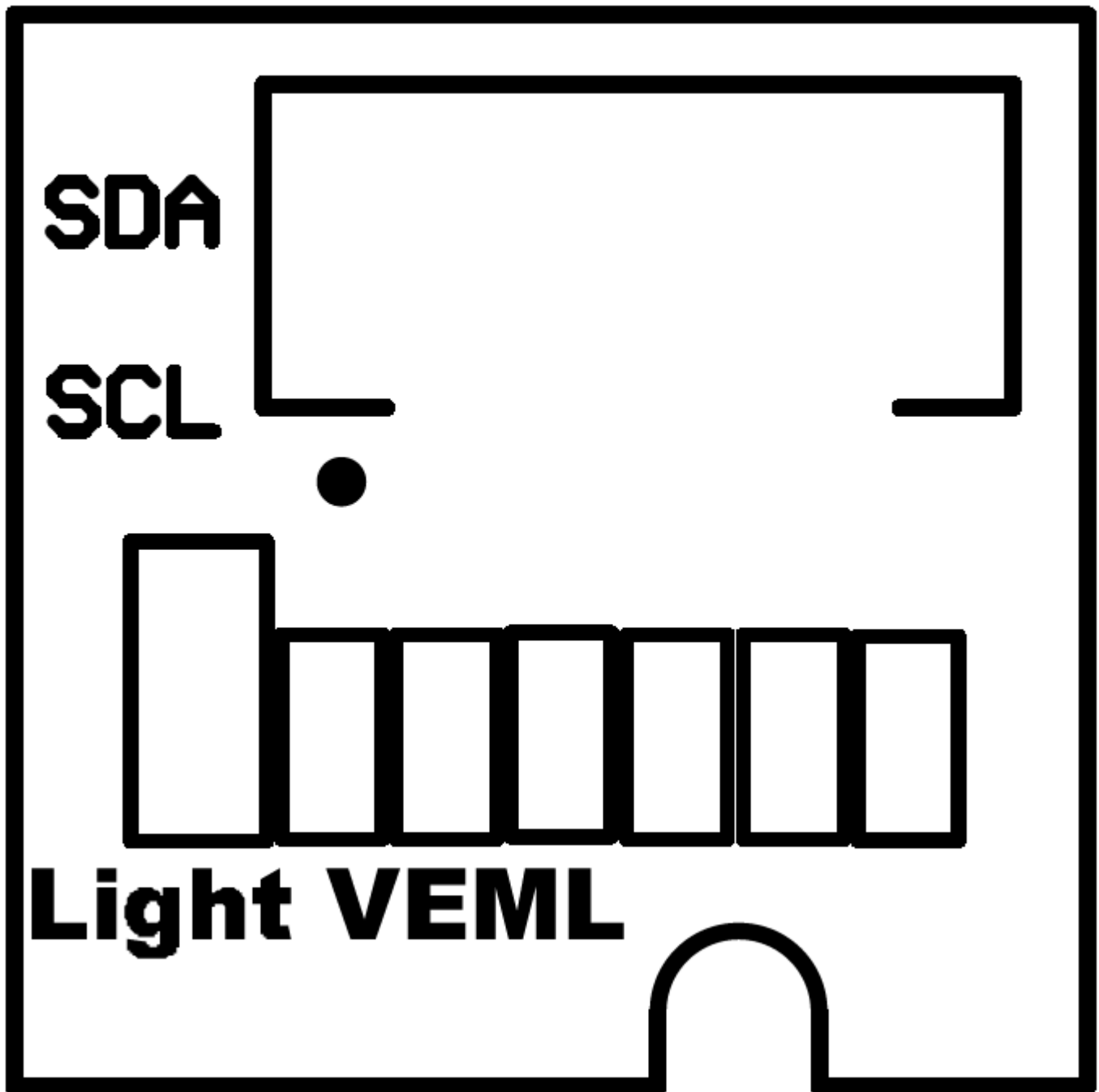


Figure 6: RAK12010 PCB Silkscreen